🌐 Module 05: Web Application Penetration Testing

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# Exercise 20: Test for Vulnerabilities in Application Deployment and Configuration using AI
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Scenario

Assessing web application deployment configurations and identifying unreferenced files or backup archives are critical steps in web vulnerability scanning. Platform configuration audits uncover server header disclosures, insecure HTTP options, outdated web server software, and misconfigured file extensions.

Generative AI platforms (such as ChatGPT) can be leveraged by security professionals to quickly generate automated shell scripts that combine multiple security scanners—such as **Nikto** (for web server configuration and file extension audits) and **Gobuster** (for brute-forcing unreferenced directories and backup files).

In this exercise, we generate and execute a custom automation bash script (`TestConfiguration.sh`) against **http://luxurytreats.com** to automate configuration scanning, directory enumeration, and report logging to extract web server version information (**Microsoft-IIS/7.5**).

Objective

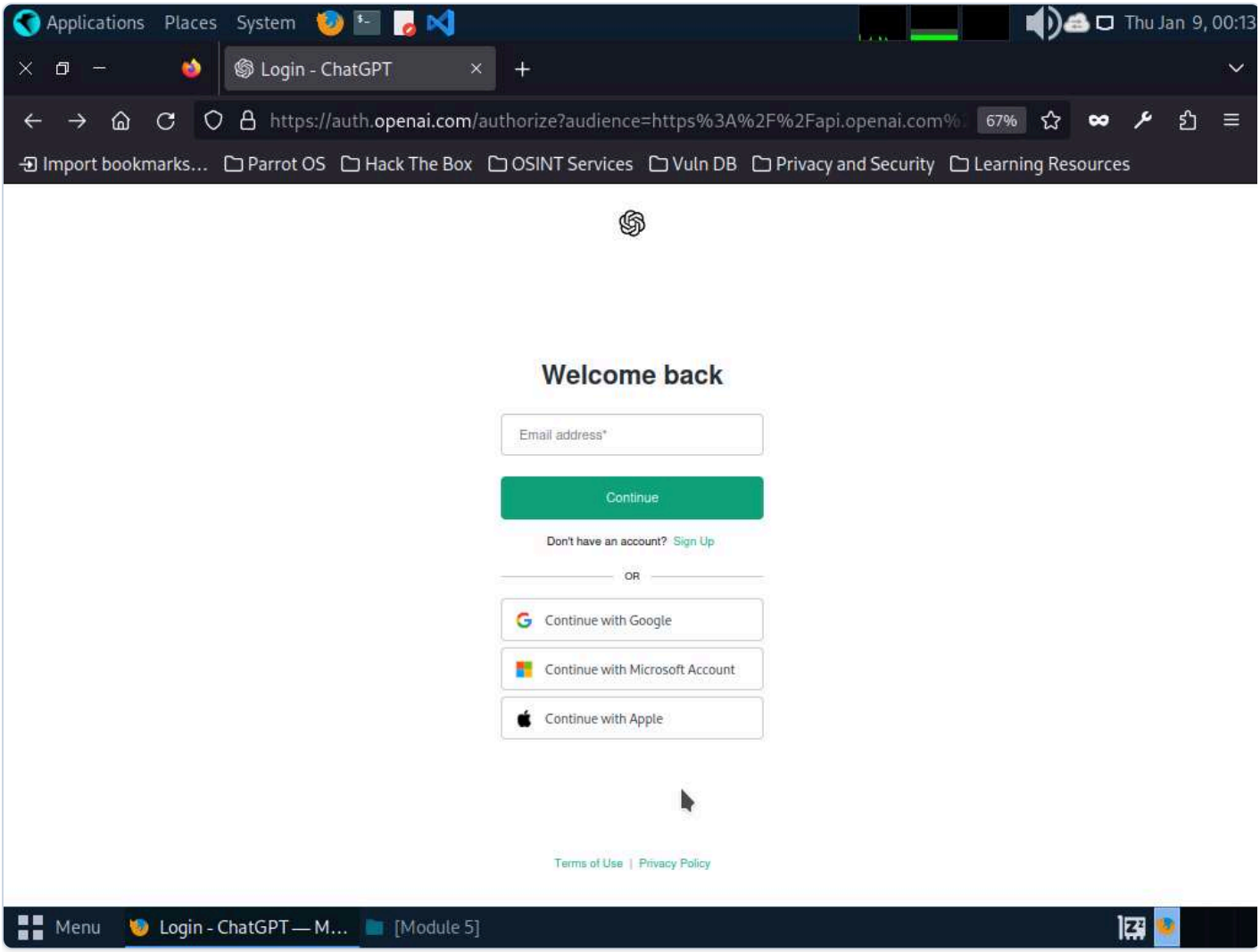
The objective of this lab is to help students learn how to:
- Prompt AI tools (ChatGPT) to generate structured shell scripts combining Nikto and Gobuster
- Prepare and convert shell scripts to Unix format (`dos2unix`) and set execution permissions (`chmod +x`)
- Perform automated web platform configuration checks and unreferenced file discovery against a target application
- Analyze audit results (`WebApp_Test_Results.txt`) to identify target web server software versions (**7.5**)

Task 1: Test for Vulnerabilities in Application Deployment and Configuration using AI

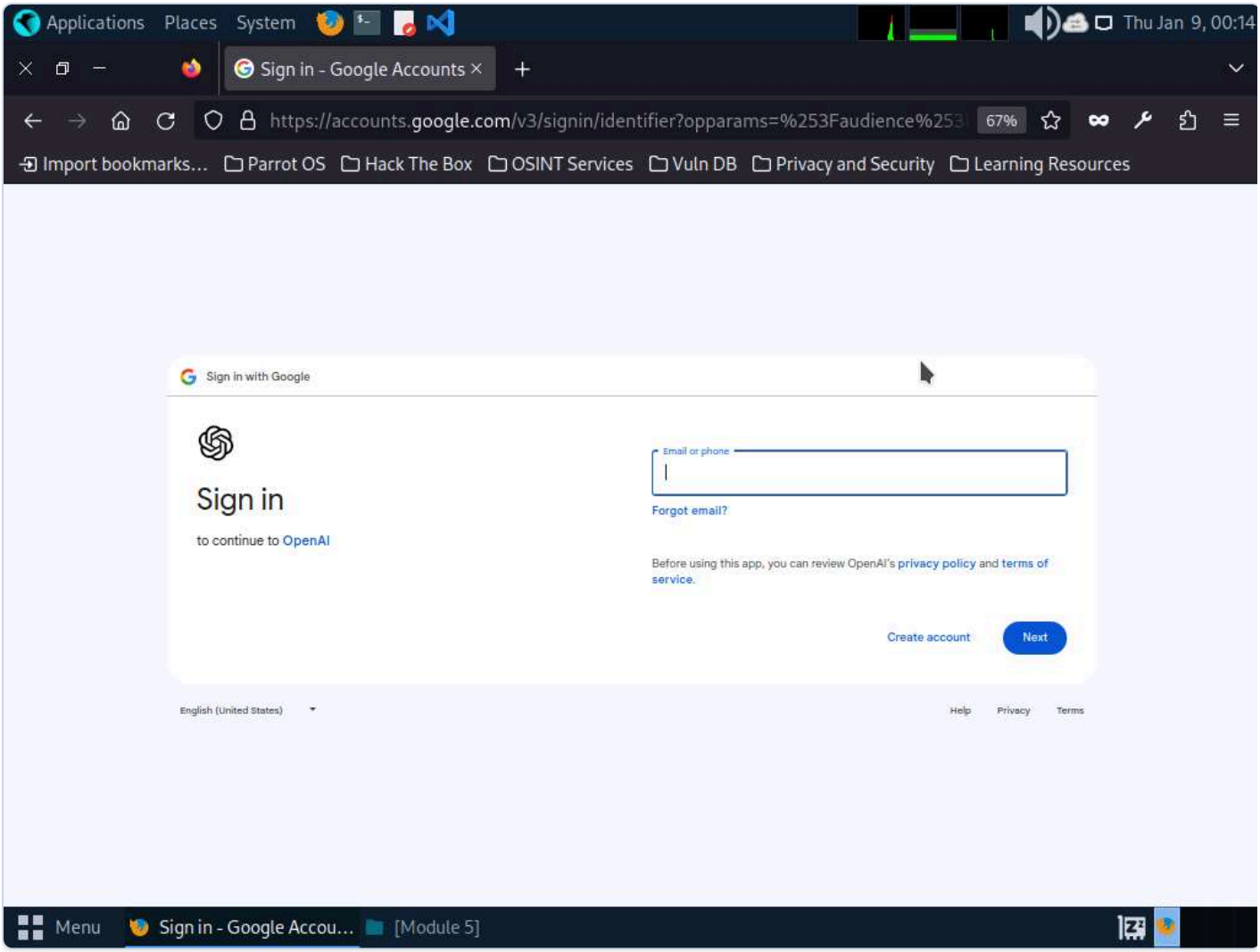
In this task, we prompt ChatGPT to craft an automated web configuration auditing script, prepare `TestConfiguration.sh` in `/home/pentester/Scripts/Module 5` , run the script against target domain **http://luxurytreats.com**, and review the generated log output (`WebApp_Test_Results.txt`) to extract the IIS server version (**7.5**).

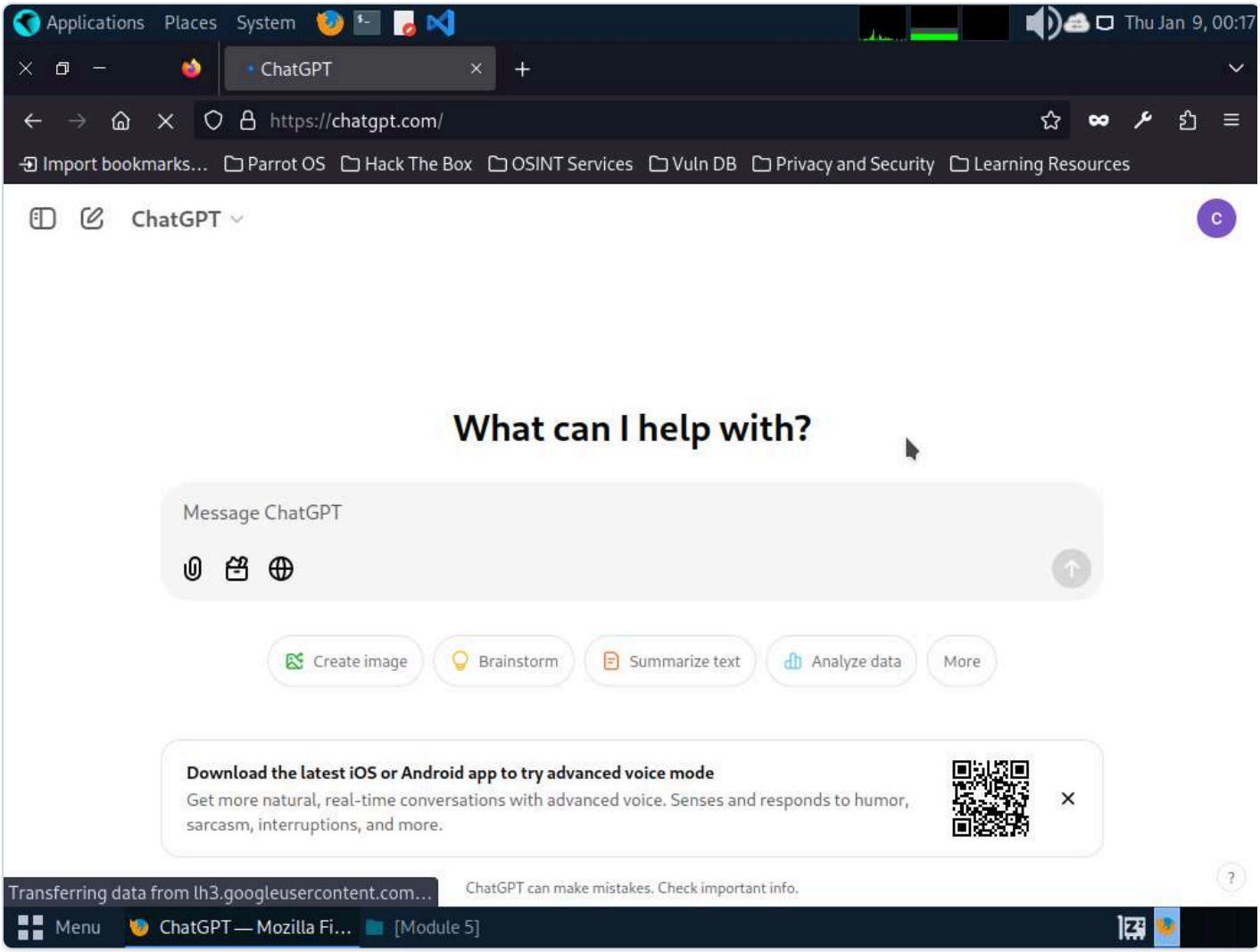
Lab Tasks

1. In **Parrot Security**, open Firefox and navigate to **https://chatgpt.com/**.



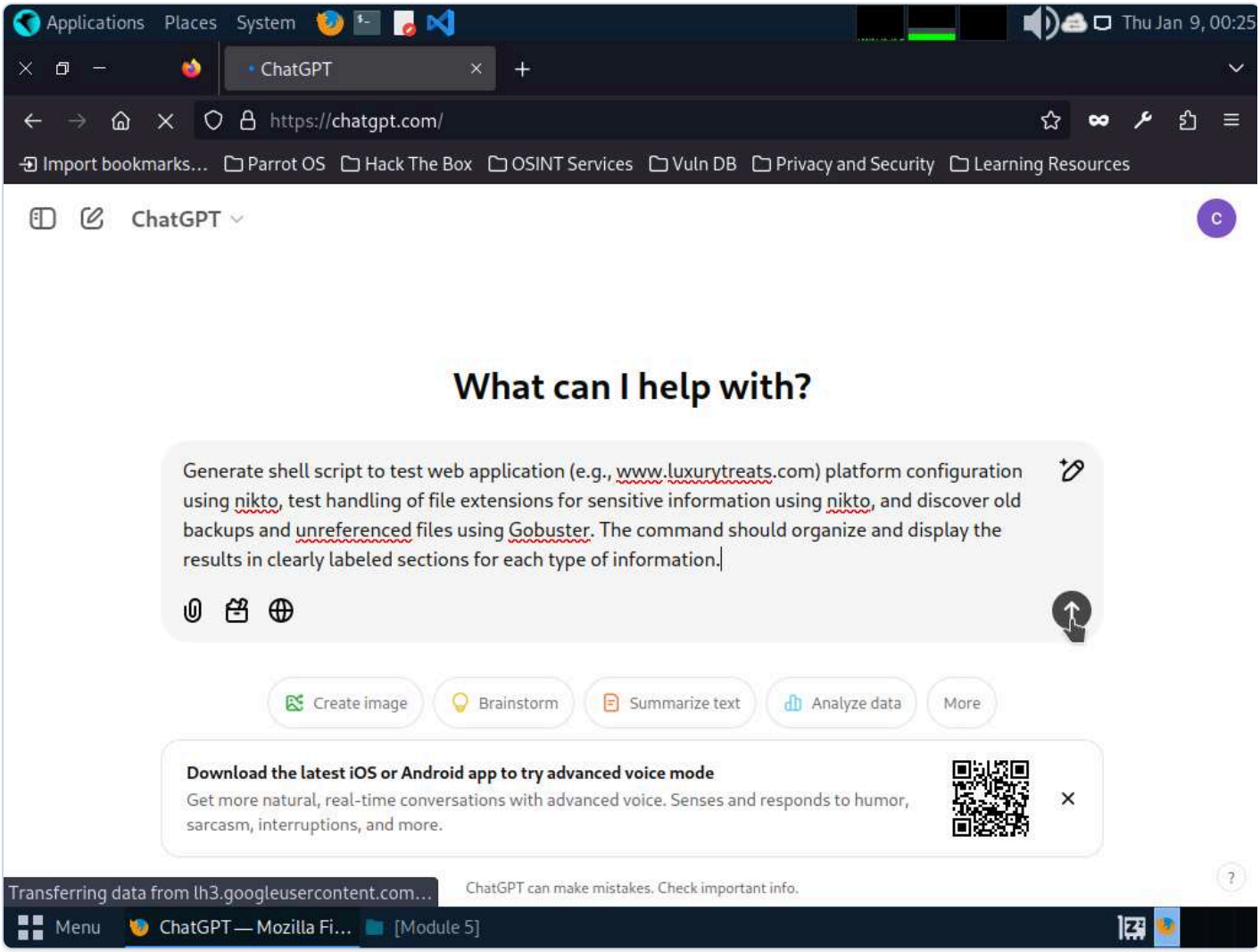
1. Log into ChatGPT using your Google or lab credentials.

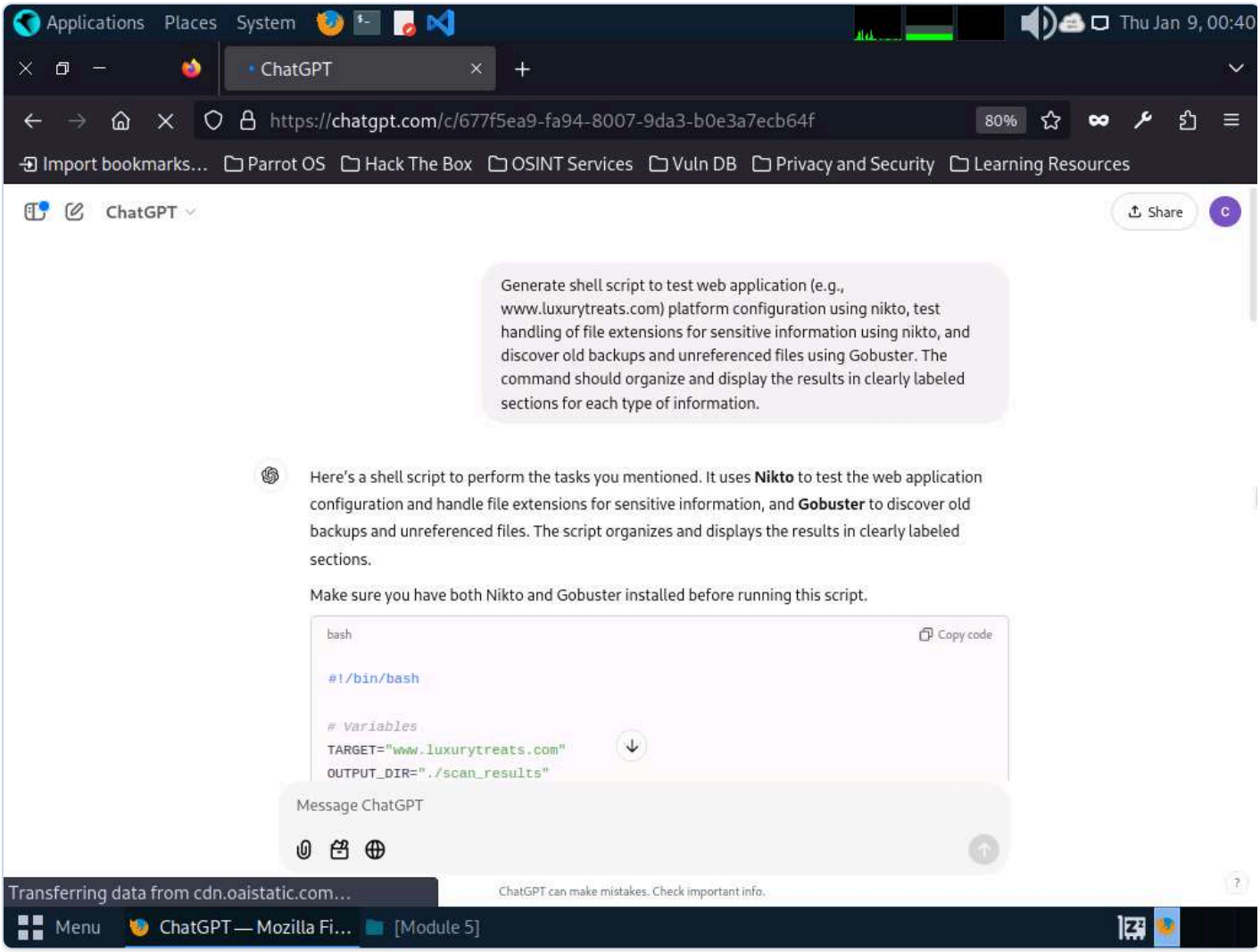

Applications
Places
System
Thu Jan 9, 00:14
Sign in - Google Accounts
https://accounts.google.com/v3/signin/identifier?opparams=%253Faudience%253
67%
Import bookmarks...
Parrot OS
Hack The Box
OSINT Services
Vuln DB
Privacy and Security
Learning Resources
Sign in with Google
Sign in
to continue to OpenAI
Email or phone
Forgot email?
Before using this app, you can review OpenAI's privacy policy and terms of service.
Create account
Next
English (United States)
Help
Privacy
Terms
Menu
Sign in - Google Accou...
[Module 5]



1. Enter the following prompt in the ChatGPT message box to generate the configuration auditing shell script:

> *"Generate shell script to test web application (e.g., www.luxurytreats.com) platform configuration using nikto, test handling of file extensions for sensitive information using nikto, and discover old backups and unreferenced files using Gobuster. The command should organize and display the results in clearly labeled sections for each type of information."*


Applications Places System
Thu Jan 9, 00:25
ChatGPT
https://chatgpt.com/
Import bookmarks...
Parrot OS
Hack The Box
OSINT Services
Vuln DB
Privacy and Security
Learning Resources
ChatGPT
C
What can I help with?
Generate shell script to test web application (e.g., www.luxurytreats.com) platform configuration using nikto, test handling of file extensions for sensitive information using nikto, and discover old backups and unreferenced files using Gobuster. The command should organize and display the results in clearly labeled sections for each type of information.
Create image
Brainstorm
Summarize text
Analyze data
More
Download the latest iOS or Android app to try advanced voice mode
Get more natural, real-time conversations with advanced voice. Senses and responds to humor, sarcasm, interruptions, and more.
Transferring data from lh3.googleusercontent.com...
ChatGPT can make mistakes. Check important info.
Menu
ChatGPT — Mozilla Fi...
[Module 5]



1. Review the generated script structure:

```
#!/bin/bash
# TestConfiguration.sh - Automated Web Application Audit Script

echo "================================================="
echo " Web Application Configuration & Reconnaissance   "
echo "================================================="

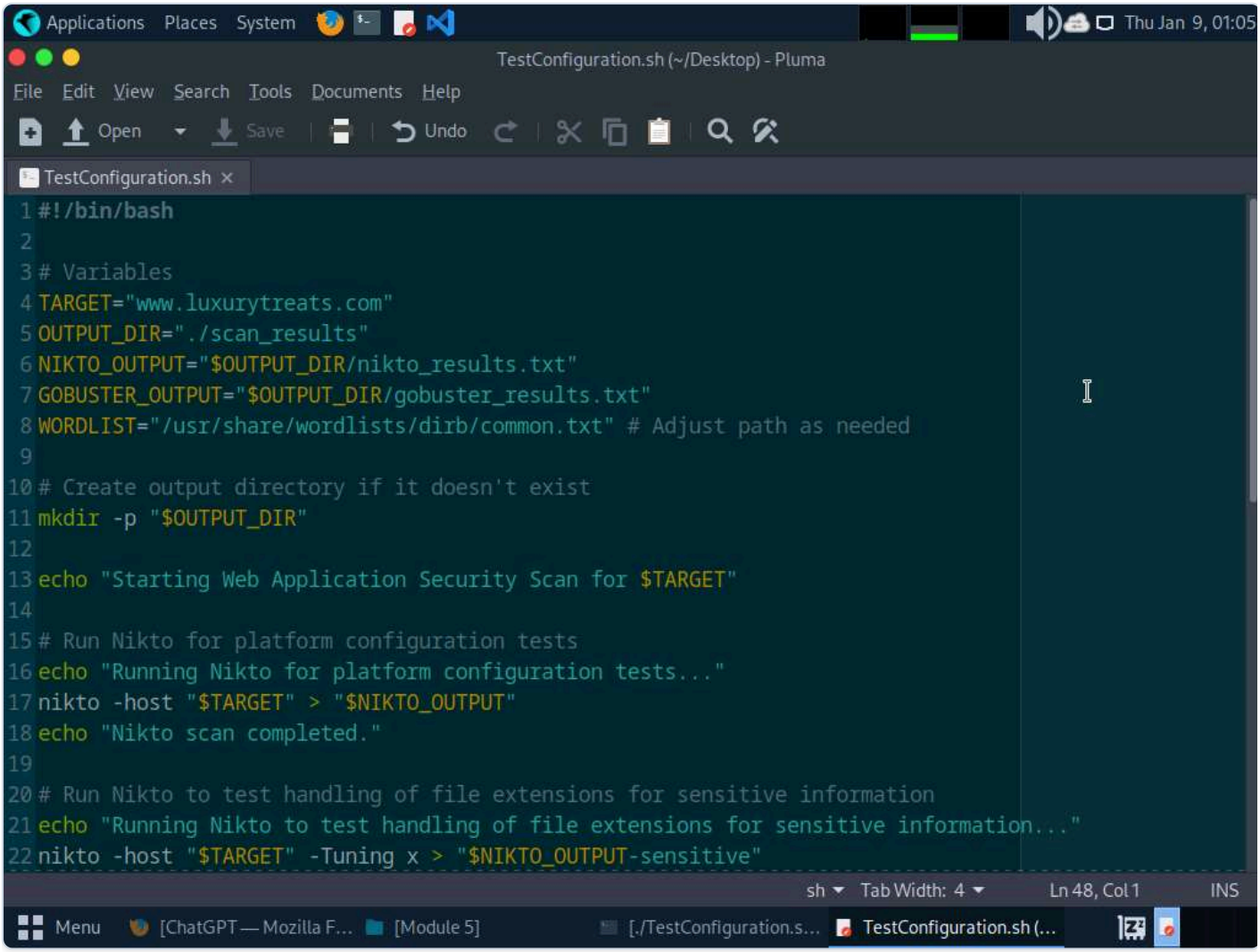
read -p "Enter Target URL (e.g., http://luxurytreats.com): " TARGET_URL
OUTPUT_FILE="WebApp_Test_Results.txt"

echo "Running Platform Configuration Scan with Nikto..." | tee -a $OUTPUT_FILE
nikto -h $TARGET_URL | tee -a $OUTPUT_FILE

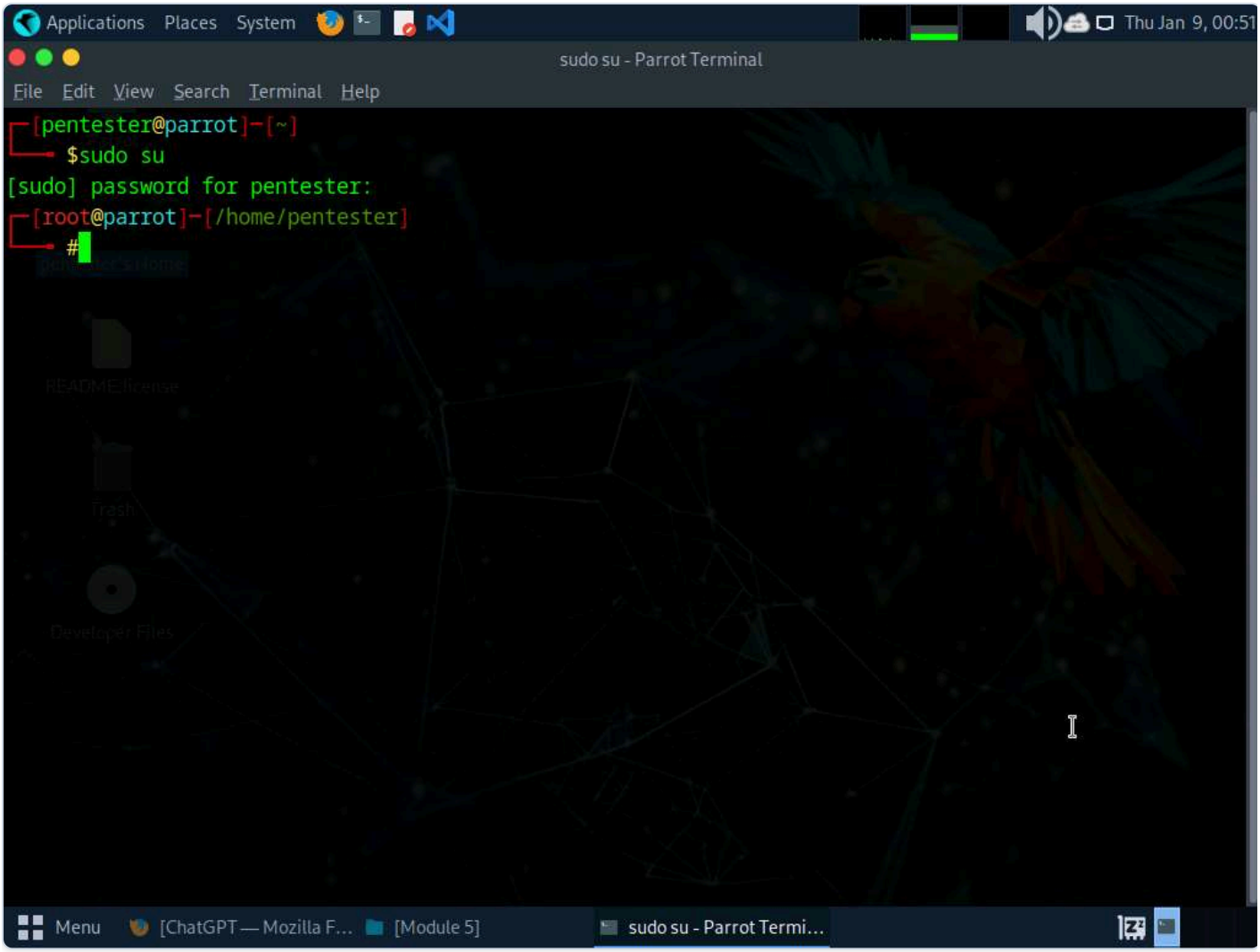
echo -e "
Running File Extension Handling Audit..." | tee -a $OUTPUT_FILE
nikto -h $TARGET_URL -Plugins "tests(file_extensions)" | tee -a $OUTPUT_FILE

echo -e "
Discovering Old Backups & Unreferenced Files with Gobuster..." | tee -a $OUTPUT_FILE
gobuster dir -u $TARGET_URL -w /usr/share/wordlists/dirb/common.txt -x php,bak,old,zip,txt | tee -a

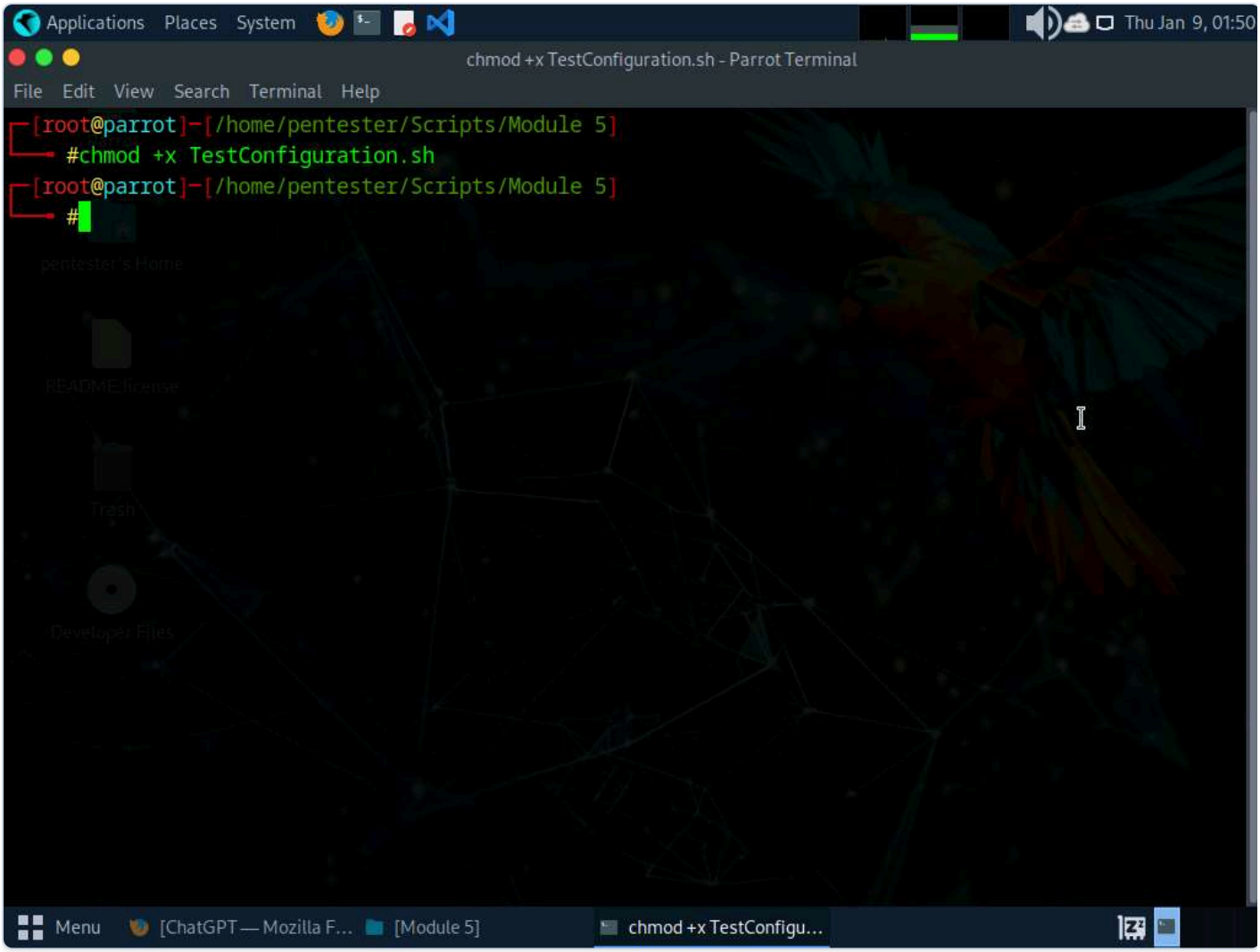
echo -e "
Audit Complete. Results saved to $OUTPUT_FILE."
```



1. Open a terminal with root privileges (**sudo su**, password: **toor**).

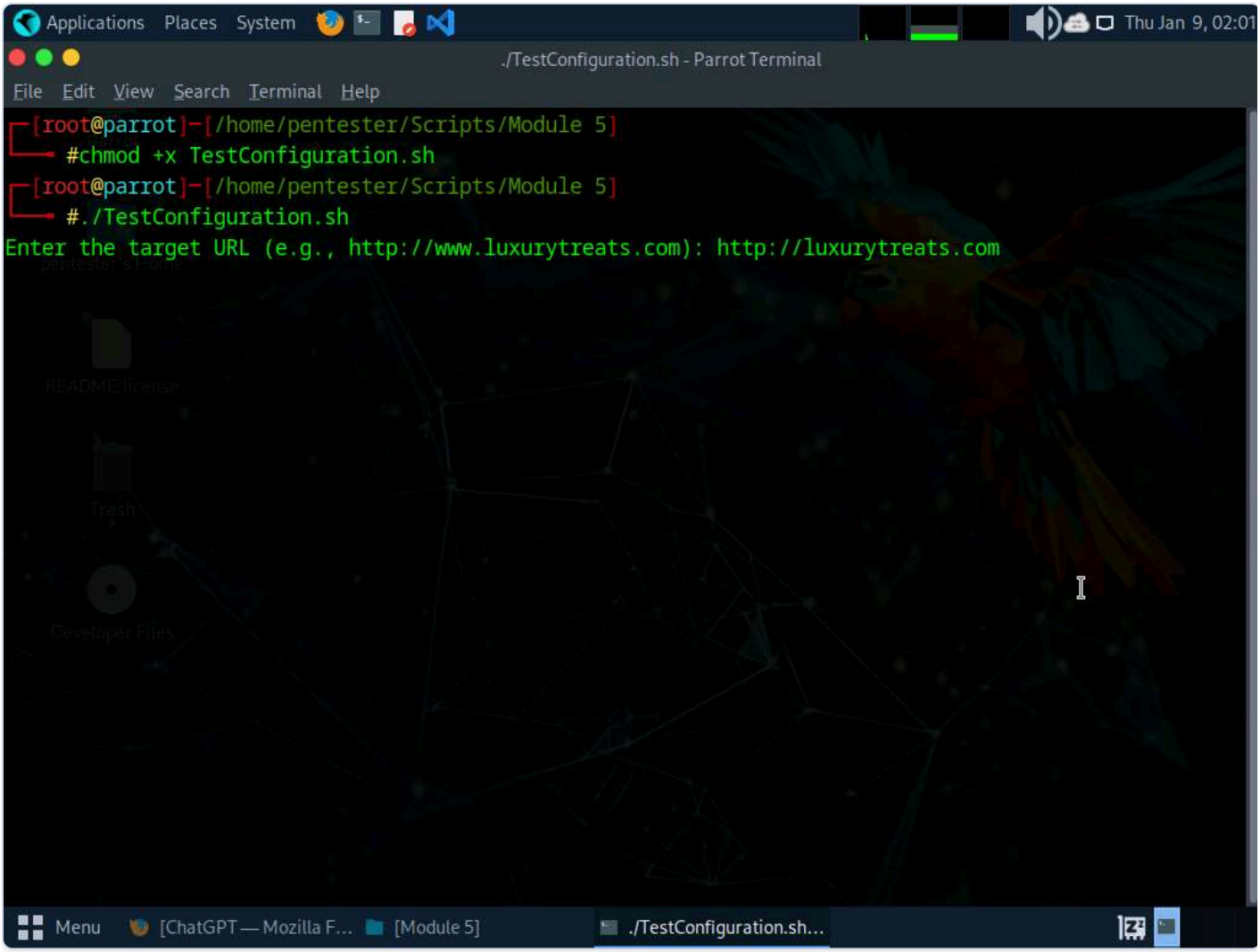


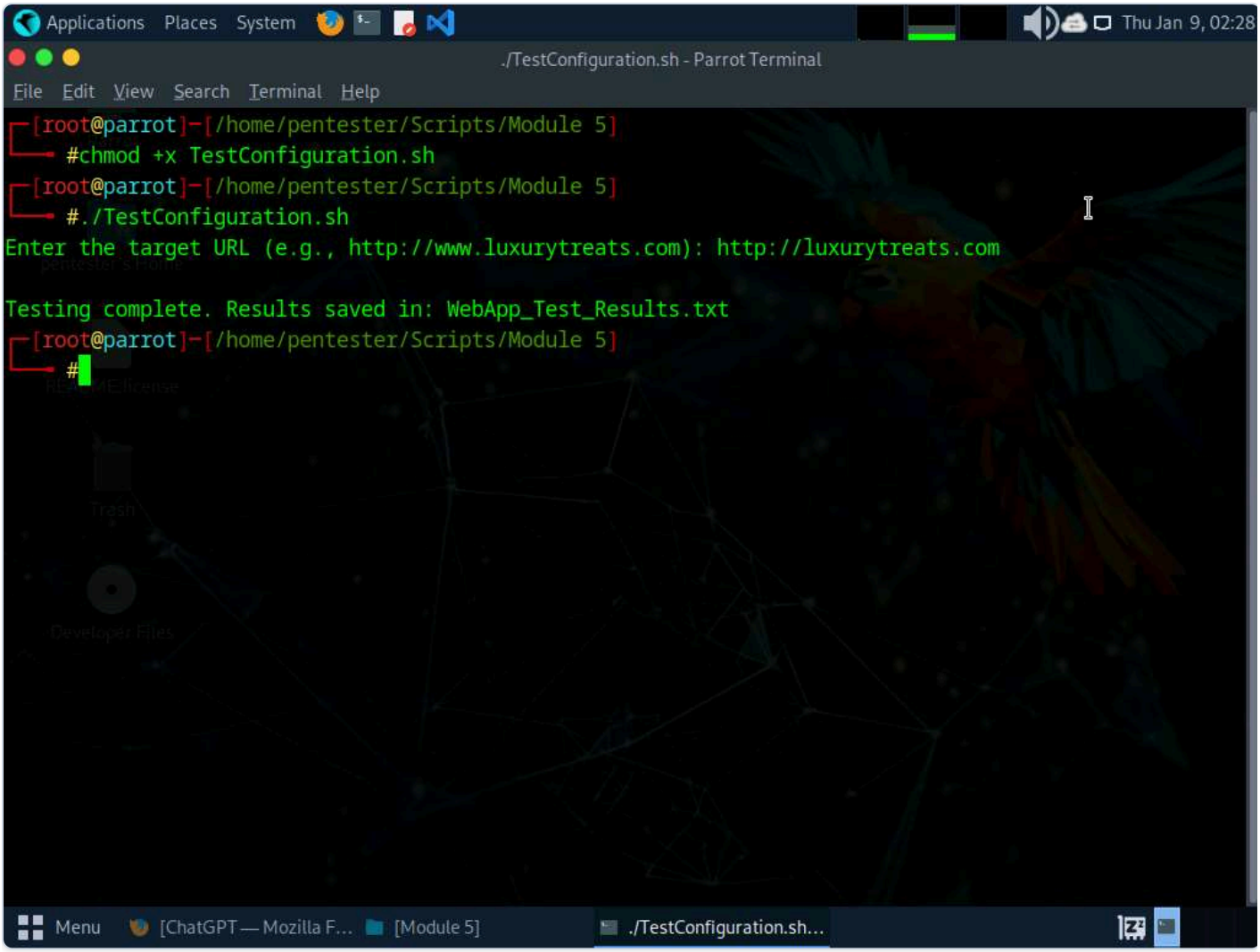
1. Navigate to the script location (**/home/pentester/Scripts/Module 5**) and make `TestConfiguration.sh` executable:
   cd "/home/pentester/Scripts/Module 5" && dos2unix TestConfiguration.sh && chmod +x TestConfiguration.sh



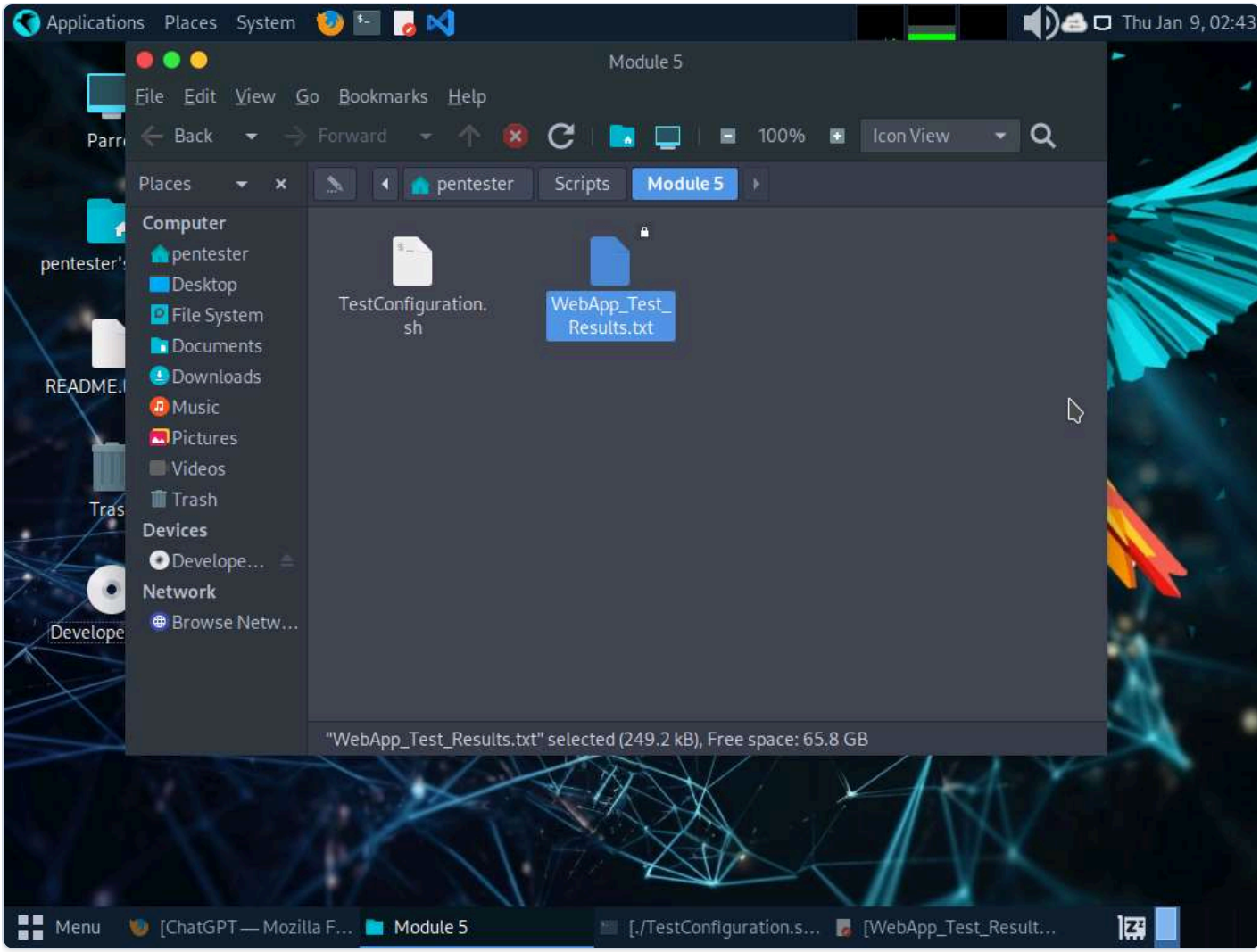
1. Execute the automation script:
   ./TestConfiguration.sh

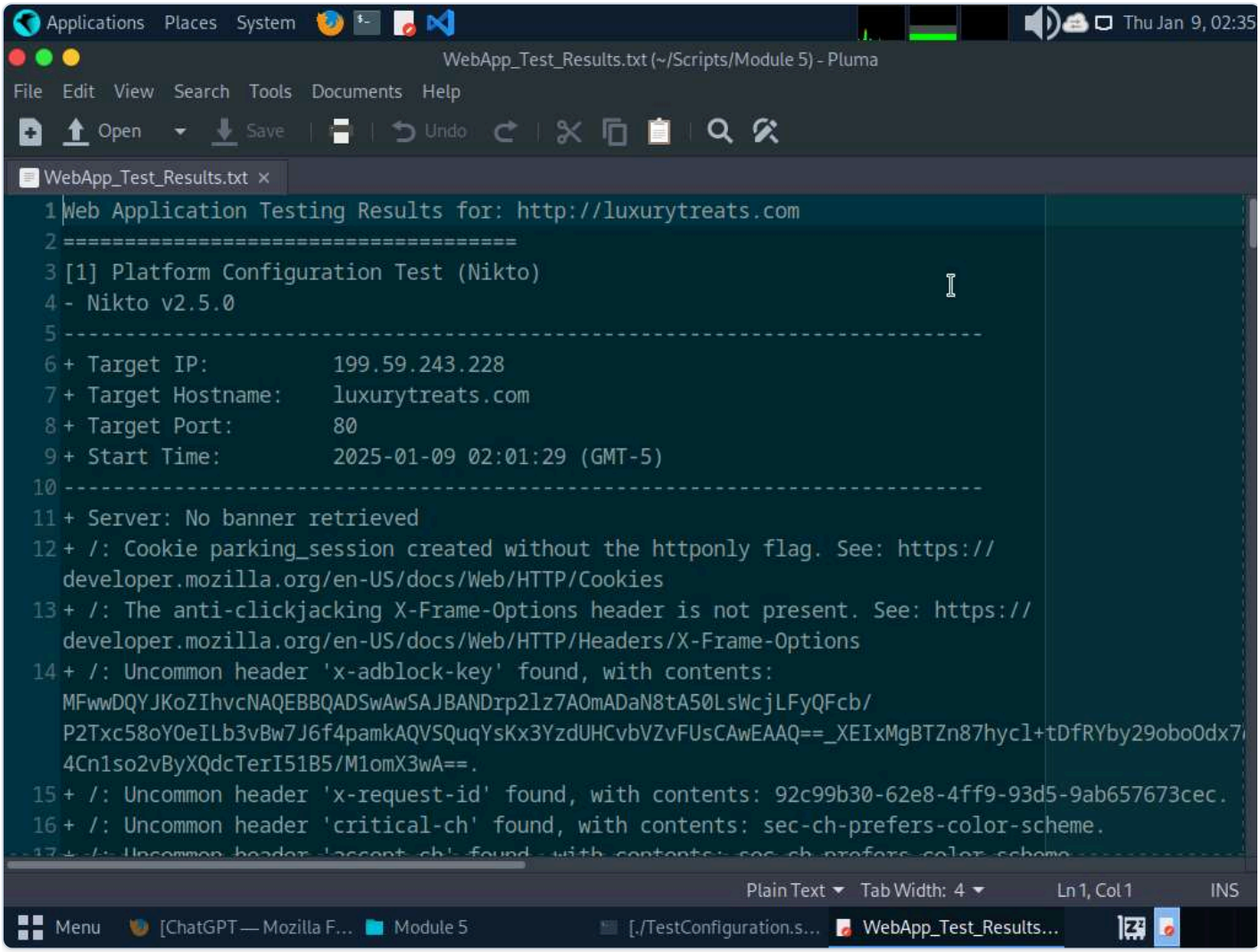
2. When prompted ("*Enter the target URL*"), enter **http://luxurytreats.com** and press Enter.

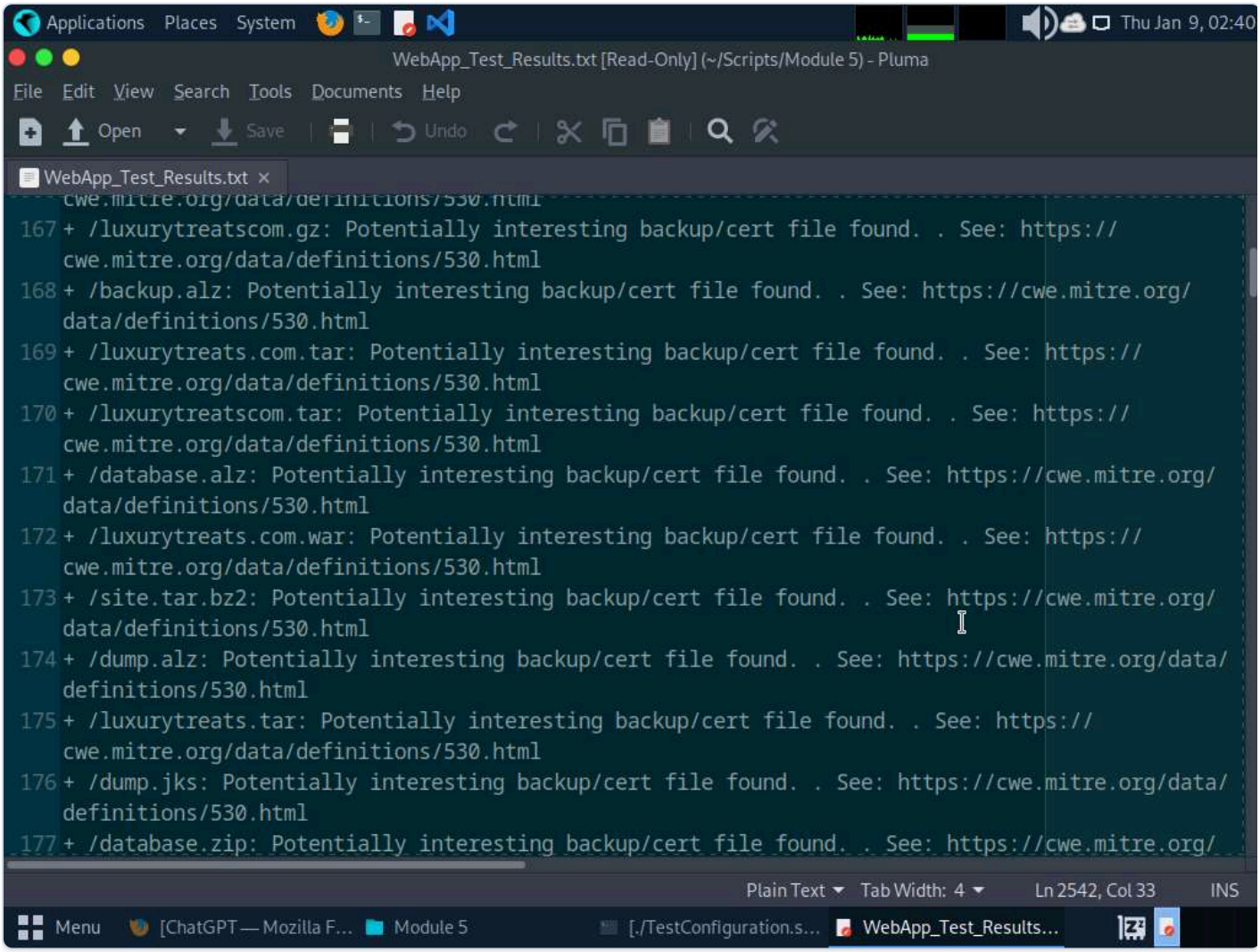

Applications Places System
Thu Jan 9, 02:01
./TestConfiguration.sh - Parrot Terminal
File Edit View Search Terminal Help
[root@parrot]-[/home/pentester/Scripts/Module 5]
#chmod +x TestConfiguration.sh
[root@parrot]-[/home/pentester/Scripts/Module 5]
#./TestConfiguration.sh
Enter the target URL (e.g., http://www.luxurytreats.com): http://luxurytreats.com
Menu
[ChatGPT — Mozilla F...
[Module 5]
./TestConfiguration.sh...



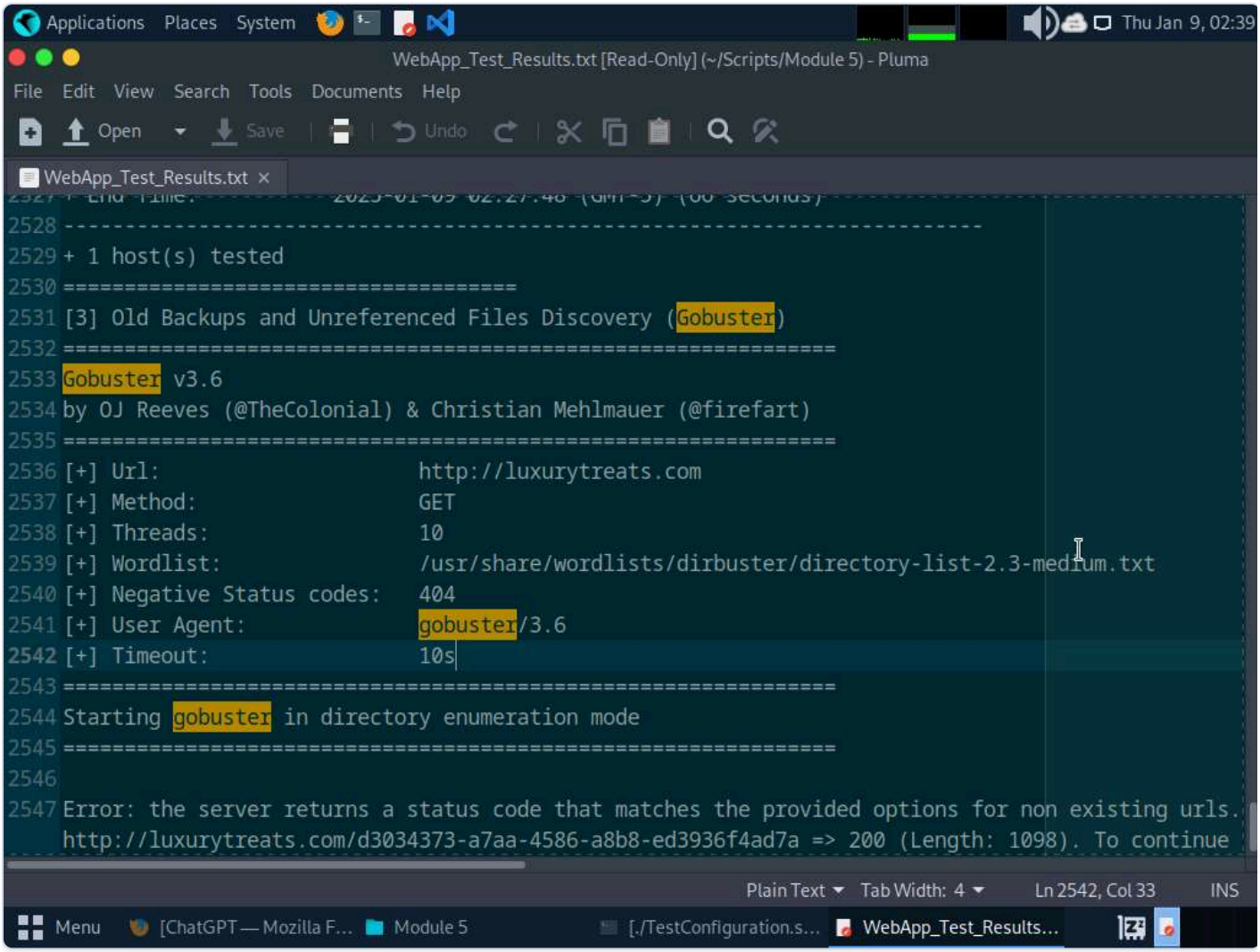
1. Wait for Nikto and Gobuster scans to complete and output to **WebApp_Test_Results.txt**.


Applications
Places
System
Thu Jan 9, 02:43
Module 5
File Edit View Go Bookmarks Help
Back
Forward
100%
Icon View
Places
pentester
Scripts
Module 5
Computer
pentester
Desktop
File System
Documents
Downloads
Music
Pictures
Videos
Trash
Devices
Develope...
Network
Browse Netw...
TestConfiguration.sh
WebApp_Test_Results.txt
"WebApp_Test_Results.txt" selected (249.2 kB), Free space: 65.8 GB
Menu
[ChatGPT — Mozilla F...
Module 5
[./TestConfiguration.s...
[WebApp_Test_Result...


Applications Places System
Thu Jan 9, 02:35
WebApp_Test_Results.txt (~/Scripts/Module 5) - Pluma
File Edit View Search Tools Documents Help
Open Save Undo
WebApp_Test_Results.txt
Web Application Testing Results for: http://luxurytreats.com
[1] Platform Configuration Test (Nikto)
- Nikto v2.5.0
+ Target IP: 199.59.243.228
+ Target Hostname: luxurytreats.com
+ Target Port: 80
+ Start Time: 2025-01-09 02:01:29 (GMT-5)
+ Server: No banner retrieved
+ /: Cookie parking_session created without the httponly flag. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Cookies
+ /: The anti-clickjacking X-Frame-Options header is not present. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Frame-Options
+ /: Uncommon header 'x-adblock-key' found, with contents: MFwwDQYJKoZIhvcNAQEBBQADSwAwSAJBANDrp2lz7AOmADaN8tA50LsWcjLFyQFcb/P2Txc58oYOeILb3vBw7J6f4pamkAQVSQuqYsKx3YzdUHCvbVZvFUsCAwEAAQ==_XEIxMgBTZn87hycl+tDfRYby29oboOdx7/4Cn1so2vByXQdcTerI51B5/M1omX3wA==.
+ /: Uncommon header 'x-request-id' found, with contents: 92c99b30-62e8-4ff9-93d5-9ab657673cec.
+ /: Uncommon header 'critical-ch' found, with contents: sec-ch-prefers-color-scheme.
Plain Text Tab Width: 4 Ln 1, Col 1 INS
Menu [ChatGPT — Mozilla F... Module 5 [./TestConfiguration.s... WebApp_Test_Results...



1. Open and review the scan results log file (**/home/pentester/Scripts/Module 5/WebApp_Test_Results.txt**).
2. Locate the **Server** response header disclosures in the Nikto output section to identify the target Microsoft IIS version: **Microsoft-IIS/7.5** (Version **7.5**).



1. Close all terminal and browser windows on Parrot Security.

This concludes the lab exercise.

> **IMPORTANT:** ❓ **Question 5.20.1.1** On a Parrot Security machine, utilize AI tools such as ChatGPT to generate a shell script that automates tasks including platform configuration checks, file extension management, and the discovery of backup files or unreferenced files on `www.luxurytreats.com`. Ensure that the script captures and records the IIS version obtained from the scan. [!SUCCESS] ✅ **Answer:** **7.5**